

Detection and Classification of AI-Generated Text

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Abstract—The advent of Large Language Models (LLMs), has largely transformed global interactions with artificial intelligence. These models, though useful in tasks ranging from translation to text generation, have now permeated everyday life, thereby raising concerns about their potential misuse, particularly in educational institutions. Despite measures like access restrictions implemented by educational institutions, AI based plagiarism has continued to grow, with existing plagiarism detection tools exhibiting limited efficacy in curbing the problem. This paper addresses the need for better AI detection algorithms, focusing on the detection of subtler forms of plagiarism facilitated by LLMs. Leveraging insights from existing studies, the research explores the efficacy of various boosting based machine learning models and dimensionality reduction techniques in improving the accuracy of detecting AI-augmented essays.

Index Terms—ChatGPT, plagiarism detection, AI in Education, Natural Language Processing

I. INTRODUCTION

ChatGPT is an advanced chatbot developed by OpenAI, an organization founded in 2015 by Elon Musk and Sam Altman, among others. Its release to the general public in November 2022 led to an unprecedented transformation at a global scale. The public went from only having a hazy idea of Artificial Intelligence (AI) to incorporating it as a part of their daily lives. This was especially true of Large Language Models (LLMs), which have become a valuable asset, serving as translators, research assistants, summarisation tools, as well as text generators.

However, in recent years, the increasing dependence on LLMs has also opened avenues for their misuse. With AI-generated research articles going undetected [1] and students being found to demonstrate increasing reliance [2] on the tool to complete their assignments, leading to concerns around gaps in education and plagiarism. To combat these issues and maintain standards of education, institutions such as New York City schools have blocked access to ChatGPT to prevent the students from using the chatbot to write their essays [3], and Seattle public schools have prohibited the use of ChatGPT [3].

Expeditious advancements in AI models have made detection of use of Large Language Models a formidable challenge [4]. Studies have shown that tools like ChatGPT are capable of producing sophisticated texts that cannot be caught by traditional plagiarism detection softwares and are difficult to

detect for the untrained eye. This poses a serious concern not only in the field of education, but also with regards to other AI-generated content like fake news and spamming, which can sway public opinion and target vulnerable communities and individuals. Not only this, as processes like answer script grading become automated [5] [6], it becomes increasingly difficult to rely on human examiners to detect the use of AI in answer scripts. It has therefore become exceedingly important to develop models that can detect the use of AI in long texts, like essays.

While models for AI detection like GPTZero exist and are able to differentiate between standard human and AI-generated text, they are unable to detect subtler methods of plagiarism. This is because, in traditional plagiarism, humans are the ones rephrasing the text, which comes with limitations on vocabulary and the amount of change they can make to the writing style. This paper therefore aims to improve the accuracy of existing models specifically on sets involving human-generated text rephrased using AI by exploring additional machine learning models and leveraging dimensionality reduction techniques.

Through this paper, we make the following contributions:

- Detection of AI-rephrased and Advanced AI-rephrased has been explored using 7 different Boosting based Algorithms.
- Different feature reduction techniques have been explored like K-best selection and Principal Component Analysis (PCA).
- Results are compared with the existing baseline models and have shown an improvement.

This paper has been divided into seven sections, with each section outlining a different stage and aspect of the study conducted. Section II is Related Work, which outlines other models that have been used to detect AI-generated essays, and analyses the pitfalls of such models. Following this, the Data Description (Section III) section is a brief on the data that was used, detailing how the data was generated and comparing other aspects of the data. Section IV, Methodology describes the boosting algorithms and dimensionality reduction techniques as well as the overall pipeline. This is

TABLE I
TABLE WITH DESCRIPTIONS AND GENERATION TECHNIQUES OF DATASETS INVOLVED^a

S. No.	Dataset Name	No. of Articles	Generated By	No. of Words	No. of Unique Words	Generation Procedure
1	Wikipedia	100	Human	38664	10513	Taken from Wikipedia on certain topics
2	Basic AI-Generated Texts	100	AI (ChatGPT)	28042	6843	3 texts for the 10 chosen categories were generated using ChatGPT, by giving the prompt: "Generate a text on the following topic: <topic>"
3	Advanced AI-Generated Texts	100	AI (ChatGPT)	25094	6166	In this case, prompt engineering was used to make the AI generate a text with the intention of it being undetectable as AI-generated. The prompt given was: "Generate a text on the following topic in a way a human would do it: <topic>"
4	Basic Rephrased Texts	100	AI (ChatGPT)	24719	7828	Wikipedia articles were rephrased for every sample using the following prompt: "Rephrase the following text: <insert Wikipedia text>"
5	Advanced AI Rephrased Texts	100	AI (ChatGPT)	20704	6413	Here, instead of simply asking the chatbot to rephrase the text, it was asked to rephrase Wikipedia text with the intent that is be unrecognizable as AI generated. The following prompt was used for this purpose: "Rephrase the following text in a way a human would do it: <insert Wikipedia text>"

^aDataset adapted from L. Mindner, T. Schlippe, and K. Schaff [7]

then followed by the Experimental Setup (Section V), which gives an overview of the platforms, tools and libraries used to conduct the experiments. The paper finally concludes with the Results (VI) and Future Scope (VII), detailing the findings of the experiments conducted, as well as laying out future avenues through which to further explore the problem at hand.

II. RELATED WORK

This section describes work on ChatGPT as well as AI detection models. It also describes certain studies that have been conducted to evaluate the extent of cheating using large language models.

Plagiarism detection has been a longstanding problem with huge consequences for the education industry, leading to constant research being done on newer more effective techniques for improving plagiarism detection [8] [9]. However, the introduction of ChatGPT changed the entire field altogether.

ChatGPT is an LLM based on Generative Pre-trained Transformers (GPT), which are pre-trained on next-word prediction tasks and then fine-tuned using downstream tasks like question answering and text classification. It is based on the GPT-3.5 architecture, which is a smaller version of GPT-3, and consists of 6.7 billion parameters, in comparison to the 175 billion parameters that GPT-3 has [10]. The advanced chatbot can perform a wide variety of tasks, and has demonstrated considerable proficiency in analytic and critical thinking related domains in higher education exams [11], as well as moderate proficiency in medical exams [12], along with being able to generate essays that are hard to detect.

According to Wulff et al. [13], a study conducted using 14 AI detection tools, including Compilatio, GPT Zero, Detect GPT, PlagiarismCheck, and Turnitin, AI paraphrased texts yielded an accuracy of only 26%, i.e. most rephrased texts went undetected. This is concerning, considering the relative

ease with which texts can be rephrased using other tools available online, thereby enabling AI use to go undetected.

Therefore, it is important to develop better algorithms to detect subtler forms of plagiarism using LLMs. Liao et al. [14] successfully detected AI-generated medical texts using Perplexity-CLS, CART, XGBoost, and BERT, to successfully arrive at an accuracy of upto 95%. Similarly, Cingillioglu and Ilker [15] were able to attain an accuracy of 92.7% using Support Vector Machines (SVMs). However, this was only done on basic essays, without any additional techniques like rephrasing being employed to evade detection.

Bhattacharjee et al. [16] similarly showed that LLMs themselves can detect AI-generated text to a considerable accuracy. Mindner et al. [7] in their paper provide additional AI feedback, perplexity based, semantic, and list lookup features that can be used to accurately detect AI-generated text up to an accuracy of 98%.

However, a limited range of algorithms for detection were explored and the results for AI rephrasing still showed a sizeable scope for further improvement.

To conclude, while considerable accuracy has been achieved in detecting simple AI-generated text in essays, accuracies for subtler evasion methods like AI-rephrased texts are much lower, leading to an increased vulnerability when it comes to plagiarism using LLMs, which can ultimately have a long lasting impact on the education system. In our study, we aim to bridge this gap, by introducing additional boosting based algorithms and leveraging feature reduction and feature subset selection methods to improve accuracy on both Basic and Advanced AI rephrased texts.

III. DATA DESCRIPTION

The dataset used for this study was generated in the paper by L. Mindner, T. Schlippe, and K. Schaff [7] The corpus

TABLE II
DESCRIPTION OF BOOSTING BASED MODELS USED

S. No.	Model Name	Model Description
1	Random Forest	Fits multiple decision tree classifiers on sub-samples of the dataset and averages them.
2	XGBoost (<i>eXtreme Gradient Boosting</i>)	Builds a predictive model by aggregating the predictions of multiple weak learners, by having each weak learner correct the mistakes of its predecessors.
3	Gradient Boosting Classifier	Fits regression trees on the negative gradient of the loss function (like binary or multiclass log loss).
4	AdaBoost (<i>Adaptive Boosting</i>)	Reassigns weights to each instance after every learning step, with greater weights being assigned to misclassified items.
5	Histogram based gradient boosting model	Decreases model training time in gradient boosting by discretizing the input data values.
6	Catboost (<i>Categorical Boosting</i>)	Handles both categorical and numerical features and uses Symmetric Weighted Quantile Sketch (SWQS) enabling it to automatically handle missing values in the dataset.
7	Light GBM (<i>Light Gradient Boosting Machine</i>)	Incorporates Gradient-based One-Side Sampling (GOSS), which selectively retains instances with large gradients during training to optimize memory usage and training time.

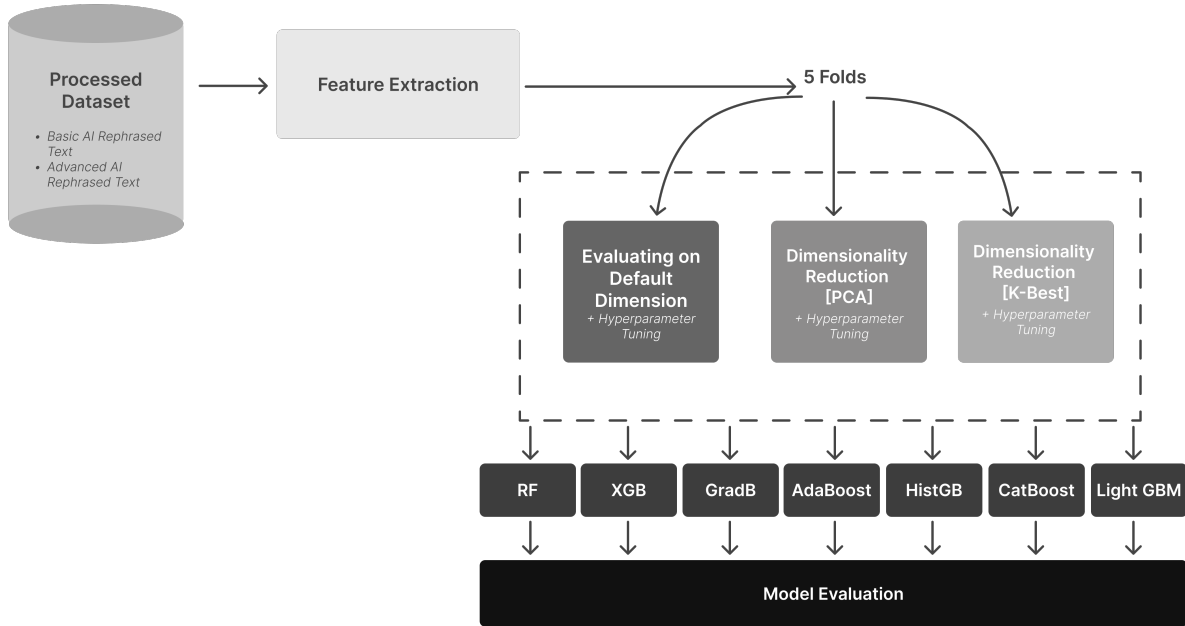


Fig. 1. Diagram of Workflow Pipeline for detection of AI-generated Paraphrased Documents

had 5 types of datasets, with 100 instances of each, which are detailed in Table I. From an initial analysis of the dataset, it is clear that the number of unique words used by humans is much greater than those used by AI, indicating key differences in the two types.

Since Mindner et al. achieved upto 98% accuracy in Basic AI-generated, and upto 97% accuracy in Advanced AI-Generated texts, this study focuses on improving the accuracy of Basic AI Rephrased texts (79%) and Advanced AI Rephrased Texts (82%), which are the last two rows of Table I. These sets were generated by asking the AI model to "rephrase the input text" and (in the case of the advanced dataset) to

"rephrase the text in the way a human would".

IV. METHODOLOGY

In this study, we expand the models explored by adding 5 new boosting-based models which are trained on the dataset. The accuracy and f1 scores of 7 different models were evaluated on 2 datasets, using the features mentioned above and adapted from the paper by Mindner et al. The accuracy of these models was compared with the baselines observed in the paper. 5-fold cross validation was performed, with a partition of 80% for training, 10% for testing, and 10% for validation.

A. Feature Extraction

Prior studies, like the one by S. Herbold, A. Hautli-Janisz, U. Heuer, Z. Kikteva, and A. Trautsch [17], have shown that certain features help better distinguish between AI-generated and human-generated texts. Complexity, and expressiveness, for example, are relatively futile features since both models depict comparable performance in both these parameters. However, there is a considerable difference in mastery of language, thereby indicating the tendency of humans to make a greater number of errors as compared to generative language models while writing essays.

Therefore, in order to incorporate these differences in features, two feature sets were extracted as described below.

- **Traditional Features:** These are standard features that are used to compare any two texts, mainly based on statistical properties of the texts being compared. The length of the feature vector here was **1035**. The features used were: *sentence bert*, *tfidf*, *flesch reading ease*, *flesch kincaid grade level*, *words per paragraph mean*, *words per paragraph SD*, *sentences per paragraph mean*, *sentences per paragraph standard deviation*, *words per sentence mean*, *words per sentence SD*, *unique words per sentence mean*, *character count*, *words count*, *words per sentence SD*, *unique words per sentence mean*, *character count*, *words count*, *unique words count*, *unique words relative*, *sentence count*, *punctuation count*, *paragraph count*, *personal pronoun relative*, *personal pronoun count*, *pos per sentence mean*, *uppercase letters relative*, *special character count*, *stop word count*, *sentiment polarity*.
- **All Features (Traditional and New):** Some additional features are included based on patterns observed in propensity of making errors, feedback received from AI models, perplexity measurements, etc., resulting in a feature vector size of **1047**. The features used here are: *Perplexity Based*, *semantic features*, *list lookup features*, *error based features*, *readability*, *AI feedback*, *text vector*, *doc and corp*.

B. Models Used

Seven boosting-based models were used. These were Random Forest (RF), XGBoost (XGB) [18], Gradient Boosting (GradB), AdaBoost [19], Histogram based boosting (HistGB) [20], Catboost [21], and Light GBM [22]. Hyperparameter tuning was also done on these models, to further improve results.

Further details on these models are given in Table II.

C. Dimensionality Reduction and Feature Subset Selection

Due to the relative recency of sophisticated language models capable of generating acceptable long-form text, finding large datasets for detection of AI-generated text is a challenge, one that the dataset considered in this study is prone to as well. This makes the task susceptible to the curse of dimensionality, taking into consideration that the feature vector

size is more than 1000 while the training set size is only 160. Therefore, it becomes imperative to use dimensionality reduction and other techniques to reduce the dimensions or the number of features being taken into consideration. In this study, a dimensionality reduction technique and a feature subset selection technique were used, Principal Component Analysis (PCA), a linear dimensionality reduction technique, and Feature Subset Selection (using k-best features). Principal Component Analysis was performed using the sklearn library and the feature set size was reduced from 1035 to around 160, corresponding to the size of the training set. On the other hand, in case of feature subset selection, the 500 best features were selected (after experimenting with different k values) from the feature set based on the *f_classif* score function, which calculates the ANOVA (Analysis of Variance) F-value between each feature and target value. The scores of the resulting models were then compared.

V. EXPERIMENTAL SETUP

This section describes the models and platforms that were used for the experiments conducted in this study. The VS Code platform was used for conducting the experiments. The sklearn library was used for the random forest model, as well as for the metrics to evaluate the models. The gradient boosted model, histogram boosted model, as well as the Adaboost classifier were also imported from sklearn. The special lightgbm (from PyPi) and catboost libraries were used for the respective models. Apart from this, the GridSearch module was used for hyperparameter tuning of each of the models. Initially, the relevant features were extracted in a separate file using the nltk library and the tfidf vectorizer from sklearn were used to tokenize the sentences and extract relevant features. Once the features were extracted, baseline results on the random forest and XGB classifier were obtained.

The models were evaluated on the basis of their F1 scores and their accuracies, after hyperparameter tuning had been performed. 5-fold cross validation, with an 80:20 split was also used to stabilize the dataset, considering a relatively smaller dataset was used. PCA and K-best selection were performed for different target feature set sizes, and a PCA size of 160 and k-best size of 500 were finally used. The feature vector was initially a size of 1035, indicating that feature reduction and subset selection techniques may prove fruitful.

VI. RESULTS

From the result tables (tables III and IV), it can be inferred that using AdaBoost gives a significant rise of almost 5% to the accuracy of the model in the case of Basic AI Rephrased text (with both traditional and all feature sets).

For the dimensionality reduction techniques, while PCA performed worse in case of all features, it performed better where a larger feature set was concerned (All features). K best, on the other hand, gave comparable results on both feature sets, indicating the redundancy of almost half the features in the feature set, and indicating the possibility of more compact detection models.

TABLE III
MODEL PERFORMANCE ON BASIC AI REPHRASED TEXT

Basic AI Rephrased Text with Traditional Features							
S. No.	Model Name	Default (Acc.)	Default (F1)	With PCA (Acc.)	With PCA (F1)	With K best (Acc.)	With K best (F1)
1	RF	0.791	0.803	0.681	0.701	0.790	0.814
2	XGB	0.830	0.847	0.786	0.762	0.766	0.772
3	GradB	0.814	0.821	0.717	0.763	0.829	0.846
4	AdaBoost	0.854	0.863	0.740	0.769	0.842	0.856
5	HistGB	0.819	0.834	0.708	0.724	0.854	0.870
6	Catboost	0.839	0.846	0.646	0.688	0.827	0.834
7	Light GBM	0.828	0.843	0.719	0.726	0.806	0.83

Basic AI Rephrased Text with All Features							
S. No.	Model Name	Default (Acc.)	Default (F1)	With PCA (Acc.)	With PCA (F1)	With K best (Acc.)	With K best (F1)
1	RF	0.698	0.708	0.810	0.818	0.736	0.742
2	XGB	0.792	0.790	0.851	0.876	0.818	0.821
3	GradB	0.818	0.816	0.806	0.808	0.818	0.816
4	AdaBoost	0.854	0.861	0.853	0.861	0.831	0.826
5	HistGB	0.828	0.828	0.874	0.897	0.807	0.806
6	Catboost	0.806	0.812	0.818	0.818	0.816	0.823
7	Light GBM	0.828	0.828	0.864	0.876	0.831	0.831

TABLE IV
MODEL PERFORMANCE ON ADVANCED AI REPHRASED TEXT

Advanced AI Rephrased Text with Traditional Features							
S. No.	Model Name	Default (Acc.)	Default (F1)	With PCA (Acc.)	With PCA (F1)	With K best (Acc.)	With K best (F1)
1	RF	0.763	0.767	0.772	0.794	0.763	0.778
2	XGB	0.824	0.842	0.836	0.848	0.775	0.777
3	GradB	0.787	0.799	0.861	0.875	0.800	0.816
4	AdaBoost	0.786	0.792	0.849	0.871	0.799	0.811
5	HistGB	0.750	0.763	0.862	0.872	0.726	0.733
6	Catboost	0.775	0.793	0.811	0.822	0.789	0.799
7	Light GBM	0.749	0.746	0.850	0.865	0.812	0.822

Advanced Rephrased Text with All Features							
S. No.	Model Name	Default (Acc.)	Default (F1)	With PCA (Acc.)	With PCA (F1)	With K best (Acc.)	With K best (F1)
1	RF	0.694	0.681	0.795	0.790	0.730	0.699
2	XGB	0.753	0.742	0.727	0.708	0.774	0.762
3	GradB	0.783	0.770	0.729	0.703	0.807	0.798
4	AdaBoost	0.830	0.824	0.751	0.755	0.820	0.811
5	HistGB	0.751	0.749	0.806	0.796	0.797	0.779
6	Catboost	0.762	0.753	0.787	0.764	0.798	0.794
7	Light GBM	0.754	0.749	0.752	0.729	0.786	0.778

TABLE V
TABLE COMPARING BASELINE AND MODEL PERFORMANCE FOR BASIC
AI REPHRASED TEXT

	Baseline (Mindner et al. 2023 [7])		Default		PCA		K-Best	
	Acc	F1	Acc	F1	Acc	F1	Acc	F1
Traditional	0.79	0.789	0.854	0.863	0.786	0.762	0.854	0.87
All	0.77	0.776	0.854	0.861	0.874	0.897	0.831	0.831

TABLE VI
TABLE COMPARING BASELINE AND MODEL PERFORMANCE FOR
ADVANCED AI REPHRASED TEXT

	Baseline (Mindner et al. 2023 [7])		Default		PCA		K-Best	
	Acc	F1	Acc	F1	Acc	F1	Acc	F1
Traditional	0.81	0.806	0.824	0.842	0.862	0.871	0.812	0.822
All	0.78	0.75	0.83	0.824	0.806	0.796	0.82	0.811

AdaBoost surpassed other classifiers for Advanced AI Rephrased text (Table IV) as well, thereby establishing it as a better suited model for this task. However, interestingly, in the traditional feature set, PCA significantly improved the model accuracy in nearly all the cases, indicating a tendency of the model to overfit while also demonstrating the repercussions of feature sets with high dimensionality.

To conclude, the results obtained indicate a sizeable improvement in scores on expanding the models to include other boosting based models (as can be seen in Table V and Table VI). The results also showed that it was favourable to reduce the data into smaller dimensions, thereby enabling quicker processing of larger datasets in the future.

VII. CONCLUSION AND FUTURE SCOPE

The study shows that the dimensions of the feature sets are greatly reducible, thereby paving the path for more compact models that perform better on larger datasets. This has tremendous applications in creating small, accessible solutions for the domain of education, where plagiarism in essays using AI models has become rampant in recent years. The study also showcases the importance of distilling models and is another example of how classical techniques like PCA still hold relevance.

There are still, however, multiple other avenues left to be explored. One such example would be generalizing the model to identify AI-generated text in more diverse pieces, including but not limited to research papers, reviews, and fiction. Other feature reduction methods like Linear Discriminant Analysis (LDA) can also be experimented with. Another important step forward will be identifying more patterns in AI-generated texts and expanding the dataset to include additional ways in which AI can be used to augment raw text data.

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